CS5187 Final Exam Review / 期末复习资料

Course / 课程: CS5187: Vision & Image
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1. Knowledge Framework / 知识框架

Part I: Low-Level Vision / 低层视觉 (lec00–lec05)

Lecture	Topic / 主题
lec00	Introduction to Computer Vision / 计算机视觉导论
lec01	Images & Image Filtering / 图像与图像滤波
lec02	Edge Detection / 边缘检测
lec03	Image Resampling & Interpolation / 图像重采样与插值
lec04	Harris Corner Detection / Harris角点检测
lec05	Feature Invariance / 特征不变性

Part II: Geometry & Appearance / 几何与外观 (lec06–lec10, lec13–lec14)

Lecture	Topic / 主题
lec06	Feature Descriptors & Matching / 特征描述子与匹配
lec07	Image Transformations & Warping / 图像变换与变形
lec08	Image Alignment / 图像对齐
lec09	RANSAC / 随机抽样一致
lec10	Cameras & Projection / 相机与投影
lec13	Binocular Stereo / 双目立体视觉
lec14	Multi-View Stereo / 多视图立体

Part III: Recognition & Deep Learning / 识别与深度学习 (lec15–lec16)

Lecture	Topic / 主题
lec15	Neural Networks / 神经网络
lec16	CNNs, RNNs, Attention & Transformers / 卷积网络、循环网络、注意力与Transformer

Part IV: Generative Models / 生成模型 (下半学期)

Lecture	Topic / 主题
lecture4	Maximum Likelihood & MLE / 最大似然估计
lecture5	Latent Variable Models & ELBO / 潜变量模型与证据下界
LVM_02	VAE & Reparameterization / 变分自编码器与重参数化
NF_01	Normalizing Flow Models (1) / 标准化流模型(1)
lecture8	Normalizing Flow Models (2) / 标准化流模型(2)
lecture9	GANs / 生成对抗网络
lecture10	f-GANs, Wasserstein GAN, BiGAN, CycleGAN / f-散度GAN、WGAN、双向GAN、循环GAN
lecture11	Energy-Based Models / 基于能量的模型
lecture12	Score Matching & NCE / 分数匹配与噪声对比估计
lecture13	Score-Based Models (1) / 基于分数的模型(1)
lecture14	Score-Based Models (2) / 基于分数的模型(2)
lecture15	Diffusion Models (DDPM, SDE) / 扩散模型
lecture16	Diffusion Models: SDE↔ODE, Architecture / 扩散模型进阶

2. Core Concepts / 核心概念梳理

2.1 Image Filtering / 图像滤波

Convolution / 卷积

- Definition (定义): Replace each pixel with a weighted sum of its neighbors using a kernel. / 用卷积核将每个像素替换为其邻域的加权和。
- Formula (公式): $G[i,j] = \sum\sum H[m,n] \cdot F[i-m, j-n]$
- Key property (关键性质): Convolution is associative and commutative; convolution with self is another Gaussian. / 卷积满足结合律和交换律。

Gaussian Filter / 高斯滤波器

- Definition (定义): A low-pass filter that removes high-frequency components. / 去除高频成分的低通滤波器。
- Property (性质): Convolving two Gaussians of σ_{1} and σ_{2} yields a Gaussian of $\sigma = \sqrt{\sigma_{1}^{2} + \sigma_{2}^{2}}$. / 两个高斯卷积仍为高斯。

Sharpening Filter / 锐化滤波器

- Definition (定义): original + (original - blurred) = $2\cdot\mathrm{original}$ - blurred. / 原图加上(原图减模糊图)实现锐化。
- Also called (亦称): High-pass filter / 高通滤波器。

2.2 Edge Detection / 边缘检测

Image Gradient / 图像梯度

- Definition (定义): $\nabla f = [\partial f/\partial x, \partial f/\partial y]$, points in direction of most rapid increase. / 指向强度增长最快的方向。
- Gradient magnitude (梯度幅值): $|\nabla f| = \sqrt{(\partial f/\partial x)^{2} + (\partial f/\partial y)^{2}}$
- Gradient direction (梯度方向): $\theta = \mathrm{atan2}(\partial f/\partial y, \partial f/\partial x)$

Sobel Operator / Sobel算子

- Definition (定义): Common approximation of derivative of Gaussian. / 高斯导数的常用近似。
- Kernels (核):
- x-direction: [[-1,0,1],[-2,0,2],[-1,0,1]]
- y-direction: [[1,2,1],[0,0,0],[-1,-2,-1]]

Canny Edge Detector / Canny边缘检测器

1. Filter image with derivative of Gaussian / 用高斯导数滤波
2. Find magnitude and orientation of gradient / 求梯度幅值和方向
3. Non-maximum suppression / 非极大值抑制
4. Hysteresis thresholding (two thresholds: high T, low t) / 滞后阈值化
5. Strong edges: R > T; Weak edges: t < R < T (kept only if connected to strong edge) / 强边缘保留，弱边缘仅在与强边缘连时保留

Scale Space / 尺度空间

- Properties (性质): Edge position may shift with σ; two edges may merge; an edge cannot split into two. / 边缘位置可能随σ偏移，可合并但不能分裂。

2.3 Image Resampling / 图像重采样

Aliasing / 混叠

- Definition (定义): Occurs when sampling rate < $2\times$ max frequency (Nyquist rate). / 采样率低于奈奎斯特率时发生。
- Solution (解决方案): Gaussian pre-filtering before subsampling. / 下采样前先做高斯预滤波。

Gaussian Pyramid / 高斯金字塔

- Definition (定义): Repeatedly blur and downsample by 2x. / 反复模糊后2倍下采样。
- Space (空间): Total space = $4/3 \times$ original image. / 总空间为原图的4/3。

Interpolation Methods / 插值方法

- Nearest-neighbor (最近邻): Fastest, but blocky. / 最快但产生方块效应。
- Bilinear (双线性): Better quality, tent function. / 质量更好。
- Bicubic (双三次): Best quality among common choices. / 常用方法中质量最好。

2.4 Harris Corner Detection / Harris角点检测

SSD Error / SSD误差

- Formula (公式): $E(u,v) = \sum_{(x,y)\in W} w(x,y) [I(x+u, y+v) - I(x,y)]^{2}$
- Small motion approximation (小运动近似): $E(u,v) \approx [u\ v] H [u; v]$, where H is the second moment matrix. / 二阶矩矩阵H近似。

Second Moment Matrix / 二阶矩矩阵

- Definition (定义): $H = \sum w(x,y) \begin{bmatrix} I_x^{2} & I_x I_y \\ I_x I_y & I_y^{2} \end{bmatrix}$
- Eigenvalues (特征值): λ_{1}, λ_{2} determine corner/edge/flat classification. / 决定角点/边缘/平坦区域分类。

Classification / 分类

Case (情况)	λ_1, λ_2	Classification (分类)
Both small (都很小)	$\lambda_1 \approx \lambda_2 \approx 0$	Flat / 平坦区域
One large, one small (一大一小)	$\lambda_1 \gg \lambda_2$ or $\lambda_2 \gg \lambda_1$	Edge / 边缘
Both large (都很大)	$\lambda_1 \sim \lambda_2 \gg 0$	Corner / 角点

Harris Operator / Harris算子

- Formula (公式): $R = \det(H) - \alpha \cdot \mathrm{trace}(H)^{2} = \lambda_{1}\lambda_{2} - \alpha(\lambda_{1}+\lambda_{2})^{2}$
- Typical α (典型参数): $\alpha \in [0.04, 0.06]$

Harris Detector Steps / Harris检测步骤

1. Compute image gradients / 计算图像梯度
2. Square of derivatives, Gaussian filter / 梯度平方后高斯滤波
3. Compute cornerness function / 计算角点响应函数
4. Non-maxima suppression / 非极大值抑制

2.5 Feature Invariance / 特征不变性

Invariance vs. Equivariance / 不变性与等变性

- Invariance (不变性): Feature locations don't change after transformation. / 变换后特征位置不变。
- Equivariance (等变性): Features detected at corresponding locations in transformed images. / 变换后特征在对应位置被检测到。

Harris Invariance Properties / Harris不变性

Transformation (变换)	Property (性质)
Translation (平移)	Equivariant / 等变
Rotation (旋转)	Equivariant (eigenvalues unchanged) / 等变
Affine intensity (仿射强度)	Partially invariant / 部分不变
Scale (尺度)	NOT invariant / 不变

Scale Invariant Detection / 尺度不变检测

- Key idea (核心思想): Find local maximum in both position AND scale. / 在位置和尺度上同时寻找局部极大值。
- Laplacian of Gaussian (LoG): Blob detector, find maxima in position-scale space. / 斑点检测器，在位置-尺度空间找极大值。
- Difference of Gaussians (DoG): Approximation of LoG, used in SIFT. / LoG的近似，SIFT中使用。

Characteristic Scale / 特征尺度

- Definition (定义): Scale producing peak LoG response. / 产生LoG峰值响应的尺度。

2.6 Feature Descriptors & Matching / 特征描述子与匹配

SIFT Descriptor / SIFT描述子

- Process (过程):
- Take 16×16 window around feature / 取16×16窗口
- Compute gradient orientation per pixel / 计算每像素梯度方向
- Threshold weak edges / 阈值化弱边缘
- Create 4×4 grid of cells, 8 orientation bins each / 4×4网格，每格8方向bin
- Result: 128-dimensional descriptor / 结果为128维描述子
- Properties (性质): Robust to viewpoint (up to ~60°), illumination changes. / 对视点变化（~60°以内）和光照变化鲁棒。

HOG Descriptor / HOG描述子

- Steps: Gradient computation \rightarrow Cell histograms (9 bins) \rightarrow Block normalization (L2-norm). / 梯度计算\rightarrow单元直方图\rightarrow块归一化。

ORB Descriptor / ORB描述子

- Steps: oFAST keypoint detection \rightarrow rBRIEF binary descriptor (256 bits). / oFAST关键点检测\rightarrowrBRIEF二进制描述子。

Feature Matching / 特征匹配

- L2 distance: $\|f_1 - f_2\|$
- Ratio distance: $\|f_1 - f_2\|$ / ‖f_{1} - f_{2}'‖ (f_{2} = best match, f_{2}' = 2nd best) / 比值距离
- ROC Curve: True positive rate vs. false positive rate; AUC summarizes performance. / ROC曲线与AUC指标。

2.7 Image Transformations / 图像变换

Transformation Hierarchy / 变换层级

Transform (变换)	Matrix (矩阵)	DOF	Properties (性质)
Translation (平移)	3x3 (last row [0,0,1])	2	Origin may shift / 原点可移动
Euclidean (欧氏)	R + t	3	Preserves distances / 保持距离
Similarity (相似)	sR + t	4	Preserves angles / 保持角度
Affine (仿射)	[A b; 0 0 1]	6	Parallel lines preserved / 平行线保持
Projective (射影/单应)	General 3x3	8	Lines preserved, parallel not / 直线保持，平行不一定

Homography / 单应变换

- Definition (定义): 8-DOF projective transformation; defined up to scale. / 8自由度射影变换，由尺度因子决定。
- Minimum matches (最少匹配): 4 point correspondences. / 4对匹配点。

Forward vs. Inverse Warping / 前向与逆向变形

- Forward warping: Send source pixels to target; may cause holes. / 源像素送到目标，可能产生空洞。
- Inverse warping: For each target pixel, sample from source via T^{-1}; use interpolation. / 对每个目标像素从源采样，需插值。

2.8 Image Alignment & Least Squares / 图像对齐与最小二乘

Least Squares / 最小二乘法

- Translation (平移): t* = mean displacement; normal equations: $A^T A \cdot t = A^T b$. / 平移的最优解为平均位移。
- Affine (仿射): 6 unknowns, need ≥ 3 matches. / 6个未知量，至少3对匹配。
- Homography (单应): Solve via SVD; h = eigenvector of $A^T A$ with smallest eigenvalue. / 通过SVD求解。

2.9 RANSAC / 随机抽样一致

Algorithm / 算法步骤

1. Randomly select s samples (minimum to fit model) / 随机选s个样本
2. Fit model / 拟合模型
3. Count inliers (within threshold ε) / 计数内点
4. Repeat N times / 重复N次
5. Choose model with most inliers / 选择内点最多的模型
6. Refit using all inliers (least squares) / 用所有内点重新拟合

Number of Rounds / 迭代次数

- Formula (公式): $N = \frac{\log(1-p)}{\log(1-(1-e)^s)}$
- p = desired probability of success / 期望成功概率
- e = outlier ratio / 外点比例
- s = minimum sample size / 最小样本数

Pros & Cons / 优缺点

- Pros: Simple, general, often works well. / 简单、通用、效果好。
- Cons: Parameters to tune; can fail for very low inlier ratios. / 需调参；内点率极低时可能失败。

2.10 Cameras & Projection / 相机与投影

Pinhole Camera / 针孔相机

- Projection (投影): (x,y,z) \rightarrow $(f \cdot x/z, f \cdot y/z)$ using similar triangles. / 利用相似三角形投影。
- Perspective projection matrix / 透视投影矩阵: `K = [[f, 0, cx], [0, f, cy], [0, 0, 1]]`

Camera Parameters / 相机参数

- Intrinsics (内参): focal length f, principal point (cx,cy), aspect ratio α, skew s. / 焦距、主点、长宽比、倾斜。
- Extrinsics (外参): Rotation R (3×3) and translation t (3×1). / 旋转矩阵和平移向量。
- Full projection (完整投影): $x_{\mathrm{pixel}} = K \left[R \mid t\right] X_{\mathrm{world}}$

Projection Types / 投影类型

- Perspective (透视): General case, parallel lines converge. / 一般情况，平行线汇聚。
- Orthographic (正交): Distance to image plane = ∞; (x,y,z) \rightarrow (x,y). / 距离无穷远。
- Weak perspective (弱透视): Scaled orthographic; good for small depth range. / 缩放正交投影。

Radial Distortion / 径向畸变

- Types: Barrel (桶形) vs. Pincushion (枕形)
- Model: $r_{\mathrm{corrected}} = r(1 + k_1 r^2 + k_2 r^4 + \cdots)$

2.11 Stereo Vision / 立体视觉

Disparity & Depth / 视差与深度

- Disparity (视差): $d = x_1 - x_2$ (horizontal pixel difference between views). / 左右视图水平像素差。
- Depth (深度): $Z = f \cdot B / d$, where B = baseline, f = focal length. / 深度与视差成反比。

Stereo Matching / 立体匹配

- SSD matching: Sum of squared differences within a window. / 窗口内平方差之和。
- Window size tradeoff: Small $\rightarrow$ more detail but more noise; Large $\rightarrow$ less noise but less detail. / 窗口大小的权衡。

Stereo as Energy Minimization / 立体视觉作为能量最小化

- Energy: $E(d) = \sum \mathrm{data\_cost}(x, d(x)) + \lambda \sum \mathrm{smoothness\_cost}(d(x), d(y))$
- Smoothness costs: Potts model (0 if same, λ if different); L1 distance. / Potts模型; L1距离。

Multi-View Stereo (MVS) / 多视图立体

- SSSD: Sum of SSD over multiple baselines. / 多基线SSD之和。
- Matching scores: SSD, SAD, ZNCC (zero-mean normalized cross-correlation). / 匹配评分方法。

Plane-Sweep Stereo / 平面扫描立体

- Sweep family of planes at different depths, reproject neighbors, compare. / 在不同深度扫描平面，重投影邻居并比较。
- NeRF: Represent scenes as $(x,y,z) \rightarrow \mathrm{RGB}+\alpha$ functions, volume rendering. / 用神经网络表示场景的辐射场。

2.12 Neural Networks / 神经网络

Linear Classifier / 线性分类器

- Score function (得分函数): $f(x, W) = Wx + b$
- Three viewpoints (三个视角): Algebraic (代数), Geometric (几何), Template matching (模板匹配)

Loss Functions / 损失函数

- Cross-entropy loss (交叉熵损失): $L_i = -\log(\exp(f_{y_i}) / \sum_j \exp(f_j))$
- Softmax (软最大): $P(y=j|x) = \exp(f_j) / \sum_k \exp(f_k)$

Neural Network Architecture / 神经网络架构

- MLP (多层感知机): Linear functions separated by non-linear activations. / 线性函数间插入非线性激活函数。
- Gradient descent (梯度下降): $W \leftarrow W - \eta \nabla L(W)$; learning rate η crucial. / 学习率η至关重要。
- Backpropagation (反向传播): Efficient gradient computation via chain rule. / 通过链式法则高效计算梯度。

2.13 CNNs, RNNs, Attention, Transformers / CNN、RNN、注意力、Transformer

Convolutional Layer / 卷积层

- Output size (输出尺寸): $(N - F)/\mathrm{stride} + 1$
- Filters (滤波器): Extend full depth of input; produce activation maps. / 滤波器覆盖输入全深度，产生激活图。
- Padding & Stride: Control output spatial dimensions. / 控制输出空间维度。

Pooling Layer / 池化层

- Max pooling (最大池化): Reduces spatial dimensions; operates per activation map. / 减小空间维度，逐激活图操作。

RNN / 循环神经网络

- Recurrence (递推): $h_t = f(W \cdot x_t + U \cdot h_{t-1})$; same weights at each time step. / 每个时间步使用相同权重。
- BPTT: Backpropagation Through Time. / 时间反向传播。
- Truncated BPTT: Backpropagate only for limited steps. / 截断时间反向传播。
- Types: Vanilla RNN, LSTM, GRU. / 类型。

Self-Attention / 自注意力

- Query-Key-Value (查询-键-值): $\mathrm{Attention}(Q,K,V) = \mathrm{softmax}(QK^T / \sqrt{d_k})V$
- Multi-Head Attention (多头注意力): Multiple attention heads in parallel; concatenate outputs. / 多个注意力头并行，拼接输出。

Transformer

- Encoder (编码器): Self-attention + FFN, with residual connections and layer norm. / 自注意力+前馈网络，带残差连接和层归一化。
- Decoder (解码器): Masked multi-head attention (cannot see future) + cross-attention + FFN. / 掩码多头注意力+交叉注意力+前馈网络。
- Positional Encoding (位置编码): Injects sequence order information. / 注入序列顺序信息。

2.14 Maximum Likelihood Estimation / 最大似然估计

KL Divergence / KL散度

- Definition (定义): $D_{\mathrm{KL}}(p|q) = \mathbb{E}_{x \sim p}[\log p(x)/q(x)] \geq 0$; asymmetric. / 非对称，衡量两个分布的差异。
- Interpretation (解释): Extra bits needed when using code optimized for q on data from p. / 用q的编码压缩来自p的数据时多用的比特数。

MLE / 最大似然估计

- Formula (公式): $\theta^* = \arg\max_\theta \sum_{x \in D} \log P_\theta(x)$
- Equivalence (等价性): Minimizing $\mathrm{KL}(p_{\mathrm{data}} | p_\theta)$ ≡ maximizing expected log-likelihood. / 最小化KL散度等价于最大化期望对数似然。

Monte Carlo Estimation / 蒙特卡洛估计

- Unbiased (无偏): $\mathbb{E}[\hat{g}] = \mathbb{E}[g(x)]$
- Variance (方差): $\mathrm{Var}(\hat{g}) = \mathrm{Var}(g(x))/T$; reduces with more samples. / 更多样本降低方差。

Bias-Variance Tradeoff / 偏差-方差权衡

- High bias (高偏差): Underfitting (model too simple). / 欠拟合。
- High variance (高方差): Overfitting (model too complex). / 过拟合。
- Regularization (正则化): $\mathrm{objective} = \mathrm{loss} + R(M)$. / 目标函数加入正则项。

2.15 Latent Variable Models & VAE / 潜变量模型与变分自编码器

Latent Variable Models / 潜变量模型

- Generative process (生成过程): $z \sim p(z)$, $x \sim p(x|z;\theta)$. / 先采样潜变量，再由其生成数据。
- Marginal likelihood (边际似然): $p(x;\theta) = \int p(x,z;\theta)\,dz$ --- often intractable. / 通常难以计算。

Mixture of Gaussians / 高斯混合模型

- $z \sim \mathrm{Categorical}(1,\dots,K)$; $p(x|z{=}k) = \mathcal{N}(\mu_k, \Sigma_k)$. / 离散潜变量的混合模型。

VAE / 变分自编码器

- Infinite mixture (无限混合): $z \sim \mathcal{N}(0,I)$; $p(x|z) = \mathcal{N}(\mu_\theta(z), \Sigma_\theta(z))$; μ_θ, Σ_θ are neural networks. / 连续潜变量的无限混合高斯。

ELBO / 证据下界

- Formula (公式): $\log p(x;\theta) \geq \sum_z q(z;\varphi) \log p(z,x;\theta) + H(q) = L(x;\theta,\varphi)$
- Equality condition (等号条件): When $q(z;\varphi) = p(z|x;\theta)$, i.e., $\mathrm{KL}(q\|p(z|x))=0$. / 当变分分布等于真实后验时等号成立。
- Decomposition (分解): $\log p(x) = \mathrm{ELBO} + D_{\mathrm{KL}}(q(z;\varphi) \| p(z|x;\theta))$

Autoencoder Perspective / 自编码器视角

- ELBO = Reconstruction term - KL term: $L = \mathbb{E}_{q_\varphi(z|x)}[\log p(x|z;\theta)] - D_{\mathrm{KL}}(q_\varphi(z|x) \| p(z))$
- Reconstruction (重建项): Encourages $\hat{x} \approx x$. / 鼓励重建接近原始输入。
- KL term (KL项): Forces latent distribution close to prior p(z). / 迫使潜变量分布接近先验。

Reparameterization Trick / 重参数化技巧

- Key idea (核心思想): Sample $\varepsilon \sim \mathcal{N}(0,I)$, then $z = \mu + \sigma\varepsilon$. / 将随机性转移到$\varepsilon$，使梯度可通过确定性函数传递。
- Benefit (优势): Enables gradient-based optimization; lower variance than REINFORCE. / 实现梯度优化，方差低于REINFORCE。

Amortized Inference / 摊销推断

- Idea (思想): Learn $q_\varphi(z|x) = \mathrm{encoder}(x)$ to map each x to variational parameters. / 学习编码器网络将x映射到变分参数，避免逐点优化。

2.16 Normalizing Flow Models / 标准化流模型

Core Idea / 核心思想

- Make $x = f_\theta(z)$ deterministic and invertible; then p(x) is tractable via change of variables. / 使映射确定且可逆，通过变量替换公式计算似然。

Change of Variables / 变量替换公式

- 1D: $p_X(x) = p_Z(h(x))|h'(x)|$, where $h = f^{-1}$. / 一维情况。
- General: $p_X(x) = p_Z(f^{-1}(x)) \cdot |\det(\partial f^{-1}/\partial x)| = p_Z(z) \cdot |\det(\partial f/\partial z)|^{-1}$. / 一般情况。

Flow of Transformations / 变换流

- Composition (组合): $z_0 \rightarrow f_1 \rightarrow z_1 \rightarrow f_2 \rightarrow \cdots \rightarrow f_M \rightarrow z_M = x$
- Likelihood: $p_X(x) = p_Z(z_0) \cdot \prod_{m=1}^{M} |\det(\partial f_m / \partial z_m)|$

Triangular Jacobian / 三角雅可比矩阵

- Key insight (关键洞察): If $x_i = f_i(z)$ depends only on $z_{\leq i}$, Jacobian is lower triangular; determinant = product of diagonals = O(n). / 三角矩阵的行列式为对角线元素之积，计算高效。

NICE / 非线性独立分量估计

- Additive coupling: $x_{1:d} = z_{1:d}$; $x_{d+1:n} = z_{d+1:n} + m_\theta(z_{1:d})$; $\det(J) = 1$ (volume-preserving). / 加性耦合层，体积保持。

Real-NVP / 实数值非体积保持

- Affine coupling: $x_{1:d} = z_{1:d}$; $x_{d+1:n} = z_{d+1:n} \odot \exp(\alpha_\theta(z_{1:d})) + \mu_\theta(z_{1:d})$; $\det(J) = \exp(\Sigma\alpha)$. / 仿射耦合层，非体积保持。

MAF vs. IAF / 掩码自回归流 vs. 逆自回归流

Model	Forward (z→x)	Inverse (x→z)	Best for
MAF	Sequential (slow)	Parallel (fast)	Density estimation / 密度估计
IAF	Parallel (fast)	Sequential (slow)	Generation / 生成

Parallel Wavenet / 并行Wavenet

- Teacher (MAF): Trained via MLE for density estimation. / 教师模型用于密度估计。
- Student (IAF): Trained via KL(student||teacher) for fast generation. / 学生模型用于快速生成。
- Result: $1000\times$ faster sampling than vanilla autoregressive. / 生成速度提升1000倍。

Gaussianization Flows / 高斯化流

- Step 1: Dimension-wise Gaussianization (inverse CDF trick). / 逐维度高斯化。
- Step 2: Rotation (orthogonal Jacobian). / 旋转变换。
- Repeat: Stack learnable Gaussian copula layers. / 堆叠可学习的高斯copula层。

2.17 Generative Adversarial Networks / 生成对抗网络

GAN Framework / GAN框架

- Generator (生成器): $z \sim p(z) \rightarrow G_\theta(z) = x$; defines implicit distribution $p_\theta(x)$. / 隐式定义数据分布。
- Discriminator (判别器): $D_\varphi(x)$ outputs probability of being real data. / 输出为真实数据的概率。
- Objective (目标): $\min_G \max_D V(D,G) = \mathbb{E}_{x \sim p_{\mathrm{data}}}[\log D(x)] + \mathbb{E}_{z \sim p_z}[\log(1-D(G(z)))]$. / 极小极大博弈。

Optimal Discriminator / 最优判别器

- $D^*(x) = \frac{p_{\mathrm{data}}(x)}{p_{\mathrm{data}}(x) + p_G(x)}$
- At optimum: $V(G^*, D^*) = 2 \cdot D_{\mathrm{JSD}}[p_{\mathrm{data}}, p_G] - \log 4$. / 最优时等价于Jensen-Shannon散度。

Jensen-Shannon Divergence / JS散度

- Properties: $D_{\mathrm{JSD}}[p,q] \geq 0$; $D_{\mathrm{JSD}}[p,q] = 0$ iff $p = q$; Symmetric. / 非负、零当且仅当相等、对称。

GAN Training Algorithm / GAN训练算法

1. Sample m real data points / 采样m个真实数据
2. Sample m noise vectors / 采样m个噪声向量
3. Update discriminator: ascend $\nabla_\varphi V$ / 更新判别器参数
4. Update generator: descend $\nabla_\theta V$ / 更新生成器参数

GAN Challenges / GAN挑战

- Mode Collapse (模式坍缩): Generator collapses to few modes. / 生成器坍缩到少数模式。
- Unstable optimization (训练不稳定): Loss oscillates; no robust stopping criteria. / 损失振荡，无可靠停止准则。
- Evaluation (评估困难): Sample quality $\neq$ likelihood. / 样本质量不等于似然。

Likelihood vs. Sample Quality / 似然与样本质量

- Case: Great likelihood, poor samples $\rightarrow$ noise mixture model. / 高似然但差样本。
- Case: Great samples, poor likelihood $\rightarrow$ memorization. / 好样本但低似然。
- Lesson: Likelihood and sample quality are not always correlated. / 似然与样本质量不一定相关。

2.18 f-Divergences & f-GANs / f-散度与f-GAN

f-Divergence / f-散度

- Definition (定义): $D_f(p, q) = \mathbb{E}_{x \sim q}\left[f\left(\frac{p(x)}{q(x)}\right)\right]$, where f is convex, lower-semicontinuous with $f(1)=0$. / 任意满足凸性、下半连续且$f(1)=0$的函数f定义的散度。
- Key property (关键性质): By Jensen's inequality, $D_f(p,q) \geq 0$; equality iff $p = q$. / 由Jensen不等式知非负，等号当且仅当两分布相同。
- Examples (示例):
- KL divergence: $f(u) = u \log u$ / KL散度
- Reverse KL: $f(u) = -\log u$ / 逆KL散度
- Jensen-Shannon: $f(u) = u\log u - (u+1)\log\frac{u+1}{2}$ / JS散度
- Total Variation: $f(u) = \frac{1}{2}|u-1|$ / 全变差距离
- Chi-squared: $f(u) = (u-1)^2$ / 卡方散度

Variational Representation of f-Divergence / f-散度的变分表示

- Fenchel Conjugate (Fenchel共轭): $f(u) = \sup_{t \in \mathrm{dom}_{f^*}} \{tu - f^*(t)\}$, where f^* is the convex conjugate. / 利用凸共轭函数表示。
- Variational lower bound (变分下界): $D_f(p,q) \geq \sup_{T} \{\mathbb{E}_{x \sim p}[T(x)] - \mathbb{E}_{x \sim q}[f^*(T(x))]\}$, where $T: \mathcal{X} \rightarrow \mathbb{R}$ is any function. / 用任意函数T给出f-散度的下界。
- Key insight (关键洞察): The density ratio p/q is unknown, but the variational form can be estimated from samples alone. / 密度比未知，但变分形式可仅从样本估计，实现似然自由训练。

f-GAN / f-GAN

- Objective (目标): $\min_\theta D_f(p_{\mathrm{data}}, p_\theta) = \min_\theta \max_\varphi \{\mathbb{E}_{x \sim p_{\mathrm{data}}}[T_\varphi(x)] - \mathbb{E}_{x \sim p_\theta}[f^*(T_\varphi(x))]\}$. / 用变分下界替代$f$-散度，实现似然自由训练。
- Training (训练): Parameterize T_φ as a neural network; optimize via min-max game. / 用神经网络参数化T，通过极小极大博弈优化。
- Recovery of original GAN (还原原始GAN): Choosing f as JS divergence recovers the original GAN objective. / 选择JS散度时退化为原始GAN目标。

2.19 Wasserstein GAN (WGAN) / Wasserstein生成对抗网络

Wasserstein Distance / Wasserstein距离

- Definition (定义): $W(p_r, p_g) = \inf_{\gamma \in \Pi(p_r, p_g)} \mathbb{E}_{(x,y) \sim \gamma}[|x - y|]$, also called Earth Mover's Distance (EMD). / 也称推土机距离，衡量将一个分布"搬运"为另一分布的最小代价。
- Properties (性质): Meaningful distance even when distributions have non-overlapping supports; continuous and differentiable w.r.t. generator parameters. / 即使分布不重叠也有意义；关于生成器参数连续可微。

Kantorovich-Rubinstein Duality / Kantorovich-Rubinstein对偶

- Dual form (对偶形式): $W(p_r, p_g) = \sup_{\|f\|_L \leq 1} \{\mathbb{E}_{x \sim p_r}[f(x)] - \mathbb{E}_{x \sim p_g}[f(x)]\}$, where $\|f\|_L \leq 1$ means f is 1-Lipschitz. / 对偶形式将Wasserstein距离转化为1-Lipschitz函数空间上的优化。
- Key advantage (关键优势): Dual form can be estimated from samples; no need to know densities. / 对偶形式可从样本估计，无需知道密度。

WGAN Training / WGAN训练

- Objective (目标): $\max_{\|D\|_L \leq 1} \{\mathbb{E}_{x \sim p_{\mathrm{data}}}[D(x)] - \mathbb{E}_{z \sim p(z)}[D(G(z))]\}$, discriminator is called "critic". / 判别器称为"评论家"，需满足1-Lipschitz约束。
- Lipschitz constraint enforcement (Lipschitz约束实现):
- Weight clipping (权重裁剪): Clamp weights to $[-c, c]$. Simple but can cause optimization issues. / 简单但可能导致优化问题。
- Gradient penalty (梯度惩罚, WGAN-GP): Add $\lambda \mathbb{E}_{\hat{x}}[(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1)^2]$ to the objective. More stable. / 更稳定的梯度惩罚方法。
- Advantages over original GAN (相比原始GAN的优势):
- Loss correlates with sample quality / 损失与样本质量相关
- More stable training / 训练更稳定
- No mode collapse in practice / 实践中无模式坍缩

2.20 BiGAN / 双向生成对抗网络

Bidirectional GAN (BiGAN) / 双向GAN

- Motivation (动机): Standard GANs learn a generator $G: z \rightarrow x$ but cannot infer the latent representation z given x. / 标准GAN只学到生成器，无法推断潜变量表示。
- Architecture (架构): Add an encoder $E: x \rightarrow z$ alongside the generator $G: z \rightarrow x$; discriminator D takes (x, z) pairs and distinguishes $(x, E(x))$ from $(G(z), z)$. / 增加编码器E，判别器在(x,z)对上区分编码对与生成对。
- Objective (目标): $\min_{G,E} \max_D V(D, E, G) = \mathbb{E}_{x \sim p_{\mathrm{data}}}[\log D(x, E(x))] + \mathbb{E}_{z \sim p(z)}[\log(1 - D(G(z), z))]$.

- Result (结果): At equilibrium, $E = G^{-1}$ (encoder inverts the generator). / 均衡时编码器为生成器的逆。
- ALi (Adversarially Learned Inference): Simultaneous proposal, equivalent to BiGAN. / 同时提出的等价方法。

2.21 CycleGAN / 循环生成对抗网络

Image-to-Image Translation / 图像到图像翻译

- Problem (问题): Given images from two domains X and Y, learn to translate between them. / 给定两个域的图像，学习在域间翻译。
- Paired vs. Unpaired (配对与非配对): Paired examples are expensive; CycleGAN works with unpaired data. / 配对数据昂贵，CycleGAN使用非配对数据。

CycleGAN Architecture / CycleGAN架构

- Generators (生成器): $G: X \rightarrow Y$ and $F: Y \rightarrow X$. / 双向生成器。
- Discriminators (判别器): D_Y distinguishes real Y from $G(X)$; D_X distinguishes real X from $F(Y)$. / 双判别器。
- Adversarial loss (对抗损失): $\mathcal{L}_{\mathrm{GAN}}(G, D_Y, X, Y) + \mathcal{L}_{\mathrm{GAN}}(F, D_X, Y, X)$. / 双向对抗损失。
- Cycle consistency loss (循环一致性损失): $\lambda(\mathbb{E}_X[\|F(G(X)) - X\|_1] + \mathbb{E}_Y[\|G(F(Y)) - Y\|_1])$. / 确保往返翻译一致。
- Overall objective (总目标): $\min_{F,G,D_X,D_Y} \mathcal{L}_{\mathrm{GAN}} + \lambda \cdot \mathcal{L}_{\mathrm{cyc}}$.

AlignFlow / 对齐流

- Key idea (核心思想): If G is a normalizing flow model, $F = G^{-1}$ is free; exactly cycle-consistent. / 若生成器为流模型，逆向自然可得，精确循环一致。
- Training (训练): Can train via MLE and/or adversarial learning. / 可通过MLE和/或对抗学习训练。

StarGAN / 星形GAN

- Extension (扩展): Handles multiple domains with a single generator. / 单一生成器处理多个域的翻译。

2.22 Energy-Based Models / 基于能量的模型

EBM Formulation / EBM公式

- Definition (定义): $p_\theta(x) = \frac{\exp(f_\theta(x))}{Z(\theta)} = \frac{1}{Z(\theta)}\exp(f_\theta(x))$, where $f_\theta(x)$ is any function (e.g., neural network) and $Z(\theta) = \int \exp(f_\theta(x))dx$ is the partition function. / 任意函数取指数后归一化，分母为配分函数。
- Key property (关键性质): Very flexible architecture — f_θ can be any neural network (no invertibility, factorization, or latent variable constraints). / 架构极其灵活，f_θ可为任意神经网络。
- Partition function (配分函数): $Z(\theta) = \int \exp(f_\theta(x))dx$ — generally intractable. / 通常不可计算。

Energy Function / 能量函数

- Interpretation (解释): $-f_\theta(x)$ is the "energy"; low energy = high probability. / 负f_θ为"能量"，低能量对应高概率。
- Non-negativity trick (非负技巧): Given any function f_θ, ensure non-negativity via $g_\theta(x) = f_\theta(x)^2$, $\exp(f_\theta(x))$, $|f_\theta(x)|$, or $\log(1+\exp(f_\theta(x)))$. / 通过平方、指数、绝对值等保证非负。

Maximum Likelihood Training / 最大似然训练

- Log-likelihood (对数似然): $\log p_\theta(x) = f_\theta(x) - \log Z(\theta)$. / 对数似然等于函数值减去配分函数对数。
- Gradient (梯度): $\nabla_\theta \log p_\theta(x_{\mathrm{train}}) = \nabla_\theta f_\theta(x_{\mathrm{train}}) - \mathbb{E}_{x \sim p_\theta}[\nabla_\theta f_\theta(x)]$. / 梯度=正相位（推高训练数据能量）减负相位（拉低模型采样能量）。
- Positive phase vs. Negative phase (正相位与负相位):
- Positive phase (正相位): $\nabla_\theta f_\theta(x_{\mathrm{train}})$ — increases probability of training data. / 提高训练数据概率。
- Negative phase (负相位): $\mathbb{E}_{x \sim p_\theta}[\nabla_\theta f_\theta(x)]$ — decreases probability of model samples. / 降低模型样本概率。

Contrastive Divergence / 对比散度

- Algorithm (算法): Instead of computing $Z(\theta)$ exactly, use Monte Carlo estimate. / 用蒙特卡洛估计代替精确计算配分函数。
- Update step (更新步骤): $\nabla_\theta f_\theta(x_{\mathrm{train}}) - \nabla_\theta f_\theta(x_{\mathrm{sample}})$, where $x_{\mathrm{sample}} \sim p_\theta$. / 用模型采样样本近似负相位。
- Key idea (核心思想): Make training data more likely than typical samples from the model. / 使训练数据比模型典型样本更高。

Sampling from EBMs / 从EBM采样

- MCMC (马尔可夫链蒙特卡洛):
- Initialize x_0 randomly / 随机初始化
- Propose $x' = x_t + \mathrm{noise}$ / 提议新样本
- If $f_\theta(x') > f_\theta(x_t)$: accept; else accept with probability $\exp(f_\theta(x') - f_\theta(x_t))$ / 接受或以一定概率接受
- Converges to p_θ, but can be slow / 收敛但可能很慢
- Langevin MCMC (朗之万MCMC): $x_{t+1} = x_t + \epsilon \nabla_x \log p_\theta(x_t) + \sqrt{2\epsilon} z_t$, where $z_t \sim \mathcal{N}(0, I)$. / 利用梯度信息加速采样的MCMC方法。
- Score function (分数函数): For EBMs, $\nabla_x \log p_\theta(x) = \nabla_x f_\theta(x)$ (since $\nabla_x \log Z(\theta) = 0$). / EBM的分数函数（对数密度对$x$的梯度）恰好等于$f_\theta$的梯度。

Score-Based Generative Modeling / 基于分数的生成模型

- Key idea (核心思想): Directly estimate the score function $\nabla_x \log p(x)$ using a neural network $s_\theta(x) \approx \nabla_x \log p(x)$. / 直接用神经网络估计分数函数。
- Score matching (分数匹配): Train s_θ by minimizing $\mathbb{E}_{p_{\mathrm{data}}}[\|s_\theta(x) - \nabla_x \log p_{\mathrm{data}}(x)\|^2]$ (or variants avoiding $\nabla_x \log p_{\mathrm{data}}$). / 通过分数匹配目标训练。
- Denoising Score Matching (去噪分数匹配): Add noise σ to data, then match the score of the perturbed distribution. / 对数据加噪后匹配扰动分布的分数。
- Noise Conditional Score Network (NCSN): Use multiple noise scales $\{\sigma_1 > \sigma_2 > \cdots > \sigma_L\}$; train one network conditioned on noise level. / 使用多尺度噪声，训练条件分数网络。
- Annealed Langevin Dynamics (退火朗之万动力学): Sequentially sample with decreasing noise levels; each level initialized from previous. / 逐步降低噪声水平采样。

EBM vs. Other Generative Models / EBM与其他生成模型比较

- vs. Autoregressive/Flow/VAE: EBM has much more flexible architecture (no restrictions on f_θ), but sampling is slower (requires MCMC). / EBM架构更灵活，但采样需要MCMC。
- vs. GAN: EBM provides stable training and tractable (unnormalized) likelihood; GAN has fast sampling but unstable training and intractable likelihood. / EBM训练稳定，GAN采样快但训练不稳定。
- Composability (可组合性): $p_{\mathrm{combined}}(x) \propto \prod_i p_i(x)^{\alpha_i}$ — can combine multiple EBMs by multiplying their densities. / 多个EBM可乘性组合，适用于零样本组合推理。

3. Key Formulas & Theorems / 重点公式与定理

3.1 Convolution / 卷积

Formula: $G[i,j] = \sum_m \sum_n H[m,n] \cdot F[i-m,j-n]$
Variables: H = kernel, F = input image, G = output image
Conditions: Kernel size determines neighborhood / 核大小决定邻域范围

3.2 Gaussian Filter / 高斯滤波

Formula: $G_\sigma(x,y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2+y^2}{2\sigma^2}\right)$
Variables: σ = standard deviation (scale) / 标准差（尺度）
Conditions: Larger σ = more blurring / σ越大越模糊

3.3 Image Gradient / 图像梯度

Formula: $\nabla f = [\partial f/\partial x, \partial f/\partial y]$; $\|\nabla f\| = \sqrt{(\partial f/\partial x)^2+(\partial f/\partial y)^2}$
Variables: $\partial f/\partial x$, $\partial f/\partial y$ = partial derivatives / 偏导数
Conditions: Computed via finite differences or derivative of Gaussian / 通过有限差分或高斯导数计算

3.4 Harris Corner Response / Harris角点响应

Formula: $R = \det(H) - \alpha \cdot \mathrm{trace}(H)^2 = \lambda_1 \lambda_2 - \alpha(\lambda_1+\lambda_2)^2$
Variables: H = second moment matrix, $\alpha \in [0.04,0.06]$, λ_1, λ_2 = eigenvalues
Conditions: $R > 0 \rightarrow$ corner; $R \approx 0 \rightarrow$ edge; $R < 0 \rightarrow$ flat (using alternate sign convention)

3.5 Perspective Projection / 透视投影

Formula: $(x,y,z) \rightarrow (fx/z + c_x, fy/z + c_y)$
Variables: f = focal length, (cx,cy) = principal point
Conditions: Pinhole camera model / 针孔相机模型

3.6 Stereo Depth / 立体深度

Formula: $Z = f \cdot B / d$
Variables: Z = depth, f = focal length, B = baseline, d = disparity
Conditions: Rectified stereo pair / 校正后的立体对

3.7 RANSAC Iterations / RANSAC迭代次数

Formula: $N = \frac{\log(1-p)}{\log(1-(1-e)^s)}$
Variables: p = success probability, e = outlier ratio, s = min sample size
Conditions: Assumes < 50% outliers / 假设外点比例 < 50%

3.8 ELBO / 证据下界

Formula: $\log p(x;\theta) \geq \mathbb{E}_q[\log p(x,z;\theta)] - \mathbb{E}_q[\log q(z;\varphi)] = \mathbb{E}_q[\log p(x|z;\theta)] - D_{\mathrm{KL}}(q_\varphi(z|x) \| p(z))$
Variables: q = variational distribution, θ = model params, φ = variational params
Conditions: Equality when $q(z;\varphi) = p(z|x;\theta)$ / 当变分分布等于后验时等号成立

3.9 Change of Variables (Flow) / 变量替换（标准化流）

Formula: $p_X(x) = p_Z(f^{-1}(x)) \cdot |\det(\partial f^{-1}/\partial x)|$
Variables: f = invertible transformation, $x,z \in \mathbb{R}^n$ (same dimension)
Conditions: f must be invertible; x,z continuous / f必须可逆；x,z连续且同维

3.10 GAN Objective / GAN目标

Formula: $\min_G \max_D \mathbb{E}_{x \sim p_{\mathrm{data}}}[\log D(x)] + \mathbb{E}_{z \sim p_z}[\log(1-D(G(z)))]$
Variables: G = generator, D = discriminator, p_z = prior on z
Conditions: At optimum, equivalent to minimizing JSD / 最优时等价于最小化JS散度

3.11 Cross-Entropy Loss / 交叉熵损失

Formula: $L_i = -\log(\exp(f_{y_i}) / \sum_j \exp(f_j))$
Variables: f_j = score for class j, y_i = true label
Conditions: Most common classification loss / 最常用的分类损失

3.12 Conv Output Size / 卷积输出尺寸

Formula: $\mathrm{output} = \frac{N - F + 2P}{\mathrm{stride}} + 1$
Variables: N = input size, F = filter size, P = padding
Conditions: Must be integer / 必须为整数

3.13 f-Divergence / f-散度

Formula: $D_f(p, q) = \mathbb{E}_{x \sim q}\left[f\left(\frac{p(x)}{q(x)}\right)\right]$, where f is convex with $f(1)=0$
Variables: p, q = probability distributions, f = convex generator function
Conditions: $D_f(p,q) \geq 0$; equals 0 iff $p = q$ / 非负，等号当且仅当两分布相同

3.14 Wasserstein Distance / Wasserstein距离

Formula: $W(p_r, p_g) = \sup_{\|f\|_L \leq 1} \mathbb{E}_{x \sim p_r}[f(x)] - \mathbb{E}_{x \sim p_g}[f(x)]$
Variables: p_r, p_g = real and generated distributions, f = 1-Lipschitz function
Conditions: Meaningful even for non-overlapping distributions / 即使分布不重叠也有意义

3.15 EBM Probability / EBM概率

Formula: $p_\theta(x) = \frac{1}{Z(\theta)}\exp(f_\theta(x))$, where $Z(\theta) = \int \exp(f_\theta(x))dx$
Variables: $f_\theta(x)$ = any function (energy negative), $Z(\theta)$ = partition function
Conditions: $Z(\theta)$ generally intractable / 配分函数通常不可计算

3.16 EBM Log-Likelihood Gradient / EBM对数似然梯度

Formula: $\nabla_\theta \log p_\theta(x_{\mathrm{train}}) = \nabla_\theta f_\theta(x_{\mathrm{train}}) - \mathbb{E}_{x \sim p_\theta}[\nabla_\theta f_\theta(x)]$
Variables: First term = positive phase, second term = negative phase
Conditions: Negative phase estimated via MCMC samples / 负相位通过MCMC样本估计

3.17 Langevin MCMC / 朗之万MCMC

Formula: $x_{t+1} = x_t + \epsilon \nabla_x f_\theta(x_t) + \sqrt{2\epsilon} z_t$, where $z_t \sim \mathcal{N}(0, I)$
Variables: ϵ = step size, $\nabla_x f_\theta(x)$ = score function
Conditions: Converges to p_θ as $\epsilon \to 0$, $T \to \infty$ / 当步长趋于0、迭代次数趋于无穷时收敛

4. Typical Questions & Solutions / 典型例题与解答

Q1: Gaussian Filter Properties / 高斯滤波器性质

Problem (问题): If you convolve an image with a Gaussian of σ_1 and then with a Gaussian of σ_2, what is the effective σ?
Solution (解答): $\sigma_{\mathrm{eff}} = \sqrt{\sigma_1^2 + \sigma_2^2}$. This follows from the property that convolution of two Gaussians is another Gaussian, and their variances add. / 两个高斯卷积仍为高斯，方差相加。

Q2: Nyquist Rate / 奈奎斯特率

Problem (问题): An image has maximum frequency 50 cycles/pixel. What is the minimum sampling rate to avoid aliasing?
Solution (解答): Nyquist rate = $2 \times$ max frequency = 100 samples/pixel interval. Sampling rate must be ≥ 100. / 采样率必须至少为最高频率的两倍。

Q3: Harris Corner Classification / Harris角点分类

Problem (问题): For a window where the second moment matrix H has eigenvalues $\lambda_1 = 5000$ and $\lambda_2 = 50$, what type of region is this?
Solution (解答): $\lambda_1 \gg \lambda_2 \rightarrow$ This is an edge region. The intensity changes significantly in one direction but not the other. / 这是一个边缘区域，强度在一个方向变化大但在另一方向变化小。

Q4: Homography DOF / 单应变换自由度

Problem (问题): How many point correspondences are needed to solve for a homography?
Solution (解答): A homography has 8 DOF (3×3 matrix up to scale). Each point correspondence gives 2 equations. Therefore, minimum 4 point correspondences are needed. / 单应矩阵8个自由度，每对点提供2个方程，至少需4对。

Q5: RANSAC Rounds / RANSAC轮数计算

Problem (问题): For computing a homography (s=4) with 30% outliers and p=0.99, how many RANSAC rounds?
Solution (解答): $N = \log(1-0.99)/\log(1-(1-0.3)^4) = \log(0.01)/\log(1-0.7^4) \approx 17$ rounds. / 约需17轮迭代。

Q6: Stereo Depth Calculation / 立体深度计算

Problem (问题): Given a stereo system with baseline B=0.2m, focal length f=500 pixels, and disparity d=50 pixels, what is the depth Z?
Solution (解答): $Z = f \cdot B / d = 500 \times 0.2 / 50 = 2.0$ meters. / 深度为2米。

Q7: Conv Layer Output Size / 卷积层输出尺寸

Problem (问题): Input $32\times32\times3$, 10 filters of size $5\times5\times3$, stride 1, padding 0. What is the output size?
Solution (解答): Output = (32-5+0)/1 + 1 = 28. So output size is $28\times28\times10$. / 输出尺寸为$28\times28\times10$。

Q8: VAE ELBO Decomposition / VAE的ELBO分解

Problem (问题): Write the ELBO in terms of reconstruction loss and KL divergence.
Solution (解答): $\mathrm{ELBO} = \mathbb{E}_{q_\varphi(z|x)}[\log p(x|z)] - D_{\mathrm{KL}}(q_\varphi(z|x) \| p(z))$ - First term: reconstruction loss (autoencoding) / 重建损失 - Second term: KL divergence of approximate posterior from prior / 近似后验与先验的KL散度

Q9: Flow Model Likelihood / 标准化流似然计算

Problem (问题): In a normalizing flow with 3 invertible layers, write the log-likelihood.
Solution (解答): $\log p_X(x) = \log p_Z(f^{-1}(x)) + \sum_{m=1}^{3} \log|\det(\partial f_m / \partial z_m)|$ where $z_0 = f^{-1}(x)$ and $z_m = f_m^{-1}(z_{m+1})$. / 对数似然等于先验对数概率加上各层雅可比行列式对数之和。

Q10: GAN Optimal Discriminator / GAN最优判别器

Problem (问题): Derive the optimal discriminator for fixed generator.
Solution (解答): $D^*(x) = \frac{p_{\mathrm{data}}(x)}{p_{\mathrm{data}}(x) + p_G(x)}$ - When $p_G = p_{\mathrm{data}}$: $D^* = 1/2$ (cannot distinguish) / 无法区分真假 - When $p_G \neq p_{\mathrm{data}}$: D* assigns higher probability to whichever distribution has higher density at x. / 判别器给密度更高的分布更高的概率。

Q11: Reparameterization Trick / 重参数化技巧

Problem (问题): Why can't we directly backpropagate through sampling z ~ q(z|x)?
Solution (解答): Sampling is a stochastic operation — gradients cannot flow through random nodes. Reparameterization: $z = \mu + \sigma\varepsilon$ where $\varepsilon \sim \mathcal{N}(0,I)$. Now z is a deterministic function of φ and ε; gradients can flow through μ and σ. / 采样是随机操作，梯度无法通过。重参数化将随机性转移到ε，使梯度可通过确定性路径μ和σ传递。

Q12: Change of Variables (1D) / 变量替换(一维)

Problem (问题): If Z ~ U[0,2] and X = 4Z, find pX(x).
Solution (解答): h(X) = X/4, h'(X) = 1/4 $p_X(x) = p_Z(X/4) \times |1/4| = (1/2) \times (1/4) = 1/8$ for $x \in [0,8]$ This is uniform on [0,8], as expected. / 结果为[0,8]上的均匀分布。

Q13: NICE vs Real-NVP / NICE与Real-NVP区别

Problem (问题): What is the key difference between NICE and Real-NVP coupling layers?
Solution (解答): - NICE: $x_{d+1:n} = z_{d+1:n} + m_\theta(z_{1:d})$; $\det(J) = 1$ (volume-preserving). / 加性耦合，体积保持。 - Real-NVP: $x_{d+1:n} = z_{d+1:n} \odot \exp(\alpha_\theta(z_{1:d})) + \mu_\theta(z_{1:d})$; $\det(J) = \exp(\sum\alpha)$ (non-volume-preserving). / 仿射耦合，非体积保持，表达能力更强。

Q14: MAF vs IAF / 掩码自回归流与逆自回归流

Problem (问题): When would you choose MAF over IAF?
Solution (解答): Choose MAF when you need efficient density evaluation (training by MLE), since its inverse $(x \rightarrow z)$ is parallel. Choose IAF when you need efficient sampling (generation), since its forward $(z \rightarrow x)$ is parallel. / MAF适合训练（密度评估快），IAF适合生成（采样快）。

Q15: Camera Projection Matrix / 相机投影矩阵

Problem (问题): Write the full projection from world coordinates to pixel coordinates.
Solution (解答): $x_{\mathrm{pixel}} = K \left[R \mid t\right] X_{\mathrm{world}}$ where $K = [[f, s, c_x], [0, \alpha f, c_y], [0, 0, 1]]$ (intrinsics), R = 3×3 rotation, t = 3×1 translation (extrinsics). / 内参矩阵K、旋转R和平移t组成完整投影。

End of CS5187 Review Material / CS5187复习资料完

Q16: WGAN vs. Original GAN / WGAN与原始GAN对比

Problem (问题): Why is Wasserstein distance preferable to Jensen-Shannon divergence as a GAN training objective?
Solution (解答): - Jensen-Shannon divergence: When p_{data} and p_G have non-overlapping supports (common in high dimensions), $D_{\mathrm{JSD}} = \log 2$ is constant — gradient provides no useful learning signal. / 当分布不重叠时JS散度为常数，无法提供学习信号。 - Wasserstein distance: Provides a meaningful, continuous gradient even when distributions don't overlap. The loss correlates with sample quality, enabling more stable training. / 即使分布不重叠也能提供有意义的连续梯度，损失与样本质量相关。 - Practical benefit: WGAN loss decreases as sample quality improves; can be used as a training monitor. / WGAN损失随样本质量提升而下降，可作为训练监控指标。

Q17: CycleGAN Cycle Consistency / CycleGAN循环一致性

Problem (问题): Why is cycle consistency necessary in CycleGAN? What happens without it?
Solution (解答): - Without cycle consistency: The adversarial losses alone only ensure that $G(X)$ looks like domain Y and $F(Y)$ looks like domain X, but there is no guarantee that $F(G(X)) \approx X$. The generators could map all inputs to a single mode in the target domain (mode collapse). / 仅对抗损失无法保证往返一致，生成器可能将所有输入映射到目标域的单一模式。 - With cycle consistency: $F(G(X)) \approx X$ and $G(F(Y)) \approx Y$ enforce that the translation preserves information content — the round trip must recover the original. / 循环一致性迫使翻译保留信息内容，往返必须恢复原始输入。 - Loss: $\mathcal{L}_{\mathrm{cyc}} = \mathbb{E}_X[\|F(G(X)) - X\|_1] + \mathbb{E}_Y[\|G(F(Y)) - Y\|_1]$. / 循环一致性损失用L1距离衡量往返差异。

Q18: EBM Gradient Interpretation / EBM梯度解释

Problem (问题): Explain the positive phase and negative phase in EBM maximum likelihood training.
Solution (解答): $\nabla_\theta \log p_\theta(x_{\mathrm{train}}) = \underbrace{\nabla_\theta f_\theta(x_{\mathrm{train}})}_{\text{positive phase}} - \underbrace{\mathbb{E}_{x \sim p_\theta}[\nabla_\theta f_\theta(x)]}_{\text{negative phase}}$ - Positive phase (正相位): Pushes up f_θ at the training data location, making training data more probable. / 在训练数据位置推高f_θ，提高训练数据概率。 - Negative phase (负相位): Pushes down f_θ at locations where the model currently assigns high probability (model samples), reducing probability of non-data regions. / 在模型当前高概率位置拉低f_θ，降低非数据区域概率。 - Contrastive divergence: Approximates the negative phase with MCMC samples: $\nabla_\theta f_\theta(x_{\mathrm{train}}) - \nabla_\theta f_\theta(x_{\mathrm{sample}})$. / 用MCMC采样近似负相位。

Q19: Score Function in EBMs / EBM中的分数函数

Problem (问题): Why is the score function $\nabla_x \log p_\theta(x)$ tractable for EBMs?
Solution (解答): $\nabla_x \log p_\theta(x) = \nabla_x [f_\theta(x) - \log Z(\theta)] = \nabla_x f_\theta(x) - 0 = \nabla_x f_\theta(x)$ Since $Z(\theta) = \int \exp(f_\theta(x))dx$ does not depend on x, its gradient w.r.t. x is zero. Therefore, the score function depends only on $f_\theta(x)$, which is easy to compute (just backprop through the network). / 配分函数不依赖于x，其对x的梯度为零，因此分数函数只依赖于$f_\theta(x)$，可通过神经网络反向传播计算。

Q20: Score Matching vs MLE / 分数匹配 vs 最大似然

Problem (问题): Why is score matching preferred over MLE for training EBMs?
Solution (解答): MLE requires computing $\log Z(\theta)$ and its gradient, which involves intractable integrals over high-dimensional spaces. Score matching sidesteps this by directly matching the score function $\nabla_x \log p_\theta(x) = \nabla_x f_\theta(x)$, which is independent of $Z(\theta)$. This avoids MCMC sampling during training. / MLE需计算不可解的配分函数，而分数匹配直接匹配分数函数（不依赖配分函数），避免了训练中的MCMC采样。

Q21: DDPM Sampling Process / DDPM采样过程

Problem (问题): Describe the generation process of a trained DDPM model.
Solution (解答): 1. Sample $x_T \sim \mathcal{N}(0,I)$ (pure noise) / 从标准高斯采样纯噪声 2. For $t = T, T-1, \ldots, 1$: / 从T到1迭代: - Predict noise: $\hat{\epsilon} = \epsilon_\theta(x_t, t)$ / 预测噪声 - Compute mean: $\mu = \frac{1}{\sqrt{1-\beta_t}}(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\hat{\epsilon})$ / 计算均值 - Sample: $x_{t-1} = \mu + \sigma_t z$, where $z \sim \mathcal{N}(0,I)$ (except at $t=1$) / 采样 3. Return x_0 / 返回最终结果

Q22: SDE vs ODE for Generation / SDE vs ODE生成

Problem (问题): Compare using the reverse SDE vs. the probability flow ODE for sampling.
Solution (解答): - Reverse SDE: Stochastic; adds noise at each step; produces diverse samples; slower convergence. / 随机性；每步加噪；产生多样样本；收敛较慢。 - Probability Flow ODE: Deterministic; unique mapping from noise to data; enables exact likelihood computation; can use adaptive step-size ODE solvers for speed. / 确定性；噪声到数据的唯一映射；可精确计算似然；可用自适应步长ODE求解器加速。 - Both have the same marginal distributions $p_t(x)$ at all times. / 两者在所有时间有相同的边际分布。

Q23: NCE vs Score Matching / NCE vs 分数匹配

Problem (问题): Compare NCE and Score Matching as training methods for EBMs.
Solution (解答):

	NCE	Score Matching
Approach / 方法	Binary classification (data vs noise) / 二分类	Minimize Fisher divergence / 最小化Fisher散度
Requires / 需要	Choice of noise distribution p_n / 选择噪声分布	Score function computation / 分数函数计算
Partition function / 配分函数	Learned as a parameter / 作为参数学习	Not needed / 不需要
Limitation / 限制	Noise must cover data support / 噪声需覆盖数据支撑集	Hessian trace (solved by DSM) / Hessian迹（DSM解决）

Part V: Score Matching & Diffusion Models / 分数匹配与扩散模型 (lecture12–16)

2.21 Score Matching / 分数匹配

Score Function / 分数函数

- Definition (定义): $s_\theta(x) = \nabla_x \log p_\theta(x)$. For EBMs, $s_\theta(x) = \nabla_x f_\theta(x)$ since the partition function $Z(\theta)$ does not depend on x. / 对数概率密度对x的梯度。EBM中分数函数与配分函数无关。
- Key Points (要点): Does not require knowledge of $Z(\theta)$; enables training without computing normalizing constants. / 无需计算归一化常数即可训练。

Fisher Divergence / Fisher散度

- Formula (公式): $D_F(p \| q) = \frac{1}{2} \mathbb{E}_{x \sim p}[\|s_p(x) - s_q(x)\|^2]$
- Key Points (要点): Measures distance between two distributions via their score functions. / 通过分数函数度量两个分布的距离。

Score Matching Objective / 分数匹配目标

- Formula (公式): $J(\theta) = \mathbb{E}_{x \sim p_{\mathrm{data}}}[\frac{1}{2}\|s_\theta(x)\|^2 + \mathrm{tr}(\nabla_x s_\theta(x))]$
- Key Points (要点): Equivalent to Fisher divergence up to a constant; requires computing Hessian trace (expensive for deep networks). / 等价于Fisher散度加常数；需计算Hessian迹。

Denoising Score Matching (DSM) / 去噪分数匹配

- Definition (定义): Match score of noise-perturbed distribution: $q_\sigma(\tilde{x}) = \int p_{\mathrm{data}}(x) \mathcal{N}(\tilde{x}|x, \sigma^2 I) dx$. / 匹配加噪扰动分布的分数。
- Formula (公式): $J_{\mathrm{DSM}} = \mathbb{E}[\|s_\theta(\tilde{x}) - (x-\tilde{x})/\sigma^2\|^2]$
- Key Points (要点): No Hessian needed; target is simply $(x - \tilde{x})/\sigma^2$; equivalent to predicting noise. / 无需Hessian；等价于预测噪声。

2.22 Noise Contrastive Estimation (NCE) / 噪声对比估计

- Definition (定义): Train discriminator to distinguish data from noise, implicitly learning the unnormalized density. / 训练判别器区分数据与噪声，隐式学习非归一化密度。
- Discriminator (判别器): $D_{\theta,Z}(x) = \frac{e^{f_\theta(x)}/Z}{e^{f_\theta(x)}/Z + p_n(x)}$
- NCE vs GAN: Both use discriminator + cross-entropy. NCE explicitly learns density ratio; GAN uses a separate generator network. / NCE显式学习密度比，GAN使用独立生成网络。

2.23 Score-Based Generative Models (NCSN) / 基于分数的生成模型

Score-Based Model / 分数模型

- Definition (定义): Neural network $s_\theta(x)$ trained to approximate $\nabla_x \log p_{\mathrm{data}}(x)$, without modeling $p_{\mathrm{data}}(x)$ itself. / 直接近似数据分布分数的神经网络。
- Sampling: Langevin MCMC: $x_{t+1} = x_t + \epsilon s_\theta(x_t) + \sqrt{2\epsilon} z_t$. / 朗之万MCMC采样。

Multiple Noise Levels (NCSN) / 多噪声水平

- Idea (思想): Train $s_\theta(x, \sigma)$ at geometric sequence of noise levels $\sigma_1 > \cdots > \sigma_L$. / 在几何递减的噪声序列上训练条件分数网络。
- Annealed Langevin Dynamics (退火朗之万): Start sampling at σ_1, gradually decrease to σ_L. / 从大噪声开始采样逐步退火。
- Key Points (要点): Large noise fills low-density regions; small noise provides details. / 大噪声填充低密度区域；小噪声提供细节。

2.24 Diffusion Models (DDPM) / 扩散模型

Forward Process / 前向过程

- Formula (公式): $q(x_t|x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} x_0, (1-\bar{\alpha}_t)I)$, $\bar{\alpha}_t = \prod_{s=1}^t (1-\beta_s)$
- Key Points (要点): Fixed, no learnable params; $x_T \approx \mathcal{N}(0,I)$ for large T. / 无可学习参数。

Reverse Process / 逆向过程

- Definition (定义): $p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sigma_t^2 I)$ / 可学习去噪。
- Training loss (训练损失): $L_{\mathrm{simple}} = \mathbb{E}_{t, x_0, \epsilon}[\|\epsilon - \epsilon_\theta(x_t, t)\|^2]$

DDPM as Hierarchical VAE / DDPM作为层次化VAE

- Encoder = fixed forward process; Decoder = learned reverse process; ELBO reduces to denoising score matching. / 编码器=固定前向过程；解码器=可学习逆向；ELBO退化为去噪分数匹配。

2.25 SDE Framework / SDE框架

Forward SDE / 前向SDE

- Formula (公式): $dx = f(x,t)dt + g(t)dw$

Reverse-Time SDE / 逆时SDE

- Formula (公式): $dx = [f(x,t) - g(t)^2 \nabla_x \log p_t(x)]dt + g(t)d\bar{w}$

Probability Flow ODE / 概率流ODE

- Formula (公式): $dx = [f(x,t) - \frac{1}{2}g(t)^2 \nabla_x \log p_t(x)]dt$
- Key Points (要点): Deterministic; same marginals as SDE; enables exact likelihood; acts as continuous normalizing flow. / 确定性；与SDE同边际；可精确计算似然。

Predictor-Corrector Sampling / 预测-校正采样

- Predictor: Numerical SDE solver (Euler-Maruyama) / 数值SDE求解器
- Corrector: Langevin MCMC step / 朗之万MCMC校正步

U-Net Architecture / U-Net架构

- Encoder-decoder with skip connections; time t injected via positional embeddings; attention at intermediate resolutions. / 带跳跃连接的编码器-解码器，时间通过位置编码注入。

Key Formulas (Supplement) / 补充公式

Denoising Score Matching / 去噪分数匹配

Formula: $J_{\mathrm{DSM}} = \mathbb{E}[\|s_\theta(\tilde{x}) - (x-\tilde{x})/\sigma^2\|^2]$
Variables: $x \sim p_{\mathrm{data}}$, $\tilde{x} = x + \sigma\epsilon$, $\epsilon \sim \mathcal{N}(0,I)$
Conditions: Small σ for good approximation / 小σ保证近似精度

DDPM Forward / DDPM前向

Formula: $q(x_t|x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)I)$
Variables: $\bar{\alpha}_t = \prod(1-\beta_s)$, $\beta_t \in [10^{-4}, 0.02]$ (linear schedule)
Conditions: $T \approx 1000$; $x_T \approx \mathcal{N}(0,I)$

DDPM Loss / DDPM损失

Formula: $L = \mathbb{E}_{t,x_0,\epsilon}[\|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t)\|^2]$
Variables: $t \sim U\{1,T\}$, $\epsilon \sim \mathcal{N}(0,I)$, ϵ_θ = U-Net
Conditions: MSE loss; equivalent to weighted ELBO / 等价于加权ELBO

Reverse SDE / 逆时SDE

Formula: $dx = [f(x,t) - g(t)^2 s_\theta(x,t)]dt + g(t)d\bar{w}$
Variables: $s_\theta \approx \nabla_x \log p_t(x)$, trained via DSM
Conditions: Continuous-time; subsumes DDPM as discrete case / 连续时间；DDPM为其离散情况